VOICE-TO-ACTION INTERLOCK · AI INFRA SUMMIT HACKATHON

The gate between hearing and hands.
Nothing moves until the command passes policy.

Bollard takes every spoken command to a robot arm, transcribes it with Speechmatics, judges it against a safety policy, and only then lets the arm move. Each command leaves a tamper-evident receipt.

Open the cockpit

See the receipts

hear → parse → judge → act → receipt · deterministic policy · no LLM in the command path



the cockpit, live idle state · two arms, one bollard, thirteen table objects

MECHANISM

01

Hear

Speechmatics Realtime streams your mic at 16 kHz. Partials arrive while you speak; the final transcript triggers judgment.

02

Judge

A deterministic grammar extracts the intent; the policy engine checks hazards, reach, grip and destination. Allow or deny, with reasons.

03

Act, then receipt

Allowed commands animate the arm on the dinner table. Every command appends a SHA-256 hash-chained receipt of what was heard, allowed, and moved.

WHAT BOLLARD ALLOWS

· place, pick up and slide safe objects: plates, bowls, cutlery, napkins

· glassware with a care note: slow approach, soft grip

· named-arm routing when you say “with the left arm”

WHAT IT REFUSES, WITH REASONS

· the hot saucepan, the lit candle, any throw of glassware

· destinations off the table edge, occupied spots, out-of-reach corners

· any command that arrives while the arm is mid-action

RECEIPTS · TWO EXAMPLES

Every command appends a receipt to a SHA-256 hash chain: the words heard, the transcription confidence, the parsed intent, the verdict with reasons, the scene fingerprint after. One click re-hashes the whole chain and says whether it holds.

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#001 · ALLOW · “place a plate on the left placemat”
Policy pass: plate ⟶ the left placemat via left arm. · scene@-0.42,0.08 · sha 8e60687e…
#002 · HAZARD_HOT · “move the pan to the center”
The saucepan is hot (60°C+ on the hotplate). Manual handling only. · sha 1ecb33fe…
```

real receipts from a live cockpit session · hit Verify chain in the app to re-hash them

Judge it yourself

built by Raphie · AI Infra Summit Hackathon, Intel online track + Speechmatics bonus

posture: simulation is policy-driven animation, not a trained VLA · STT runs on Speechmatics cloud · no LLM in the command path